



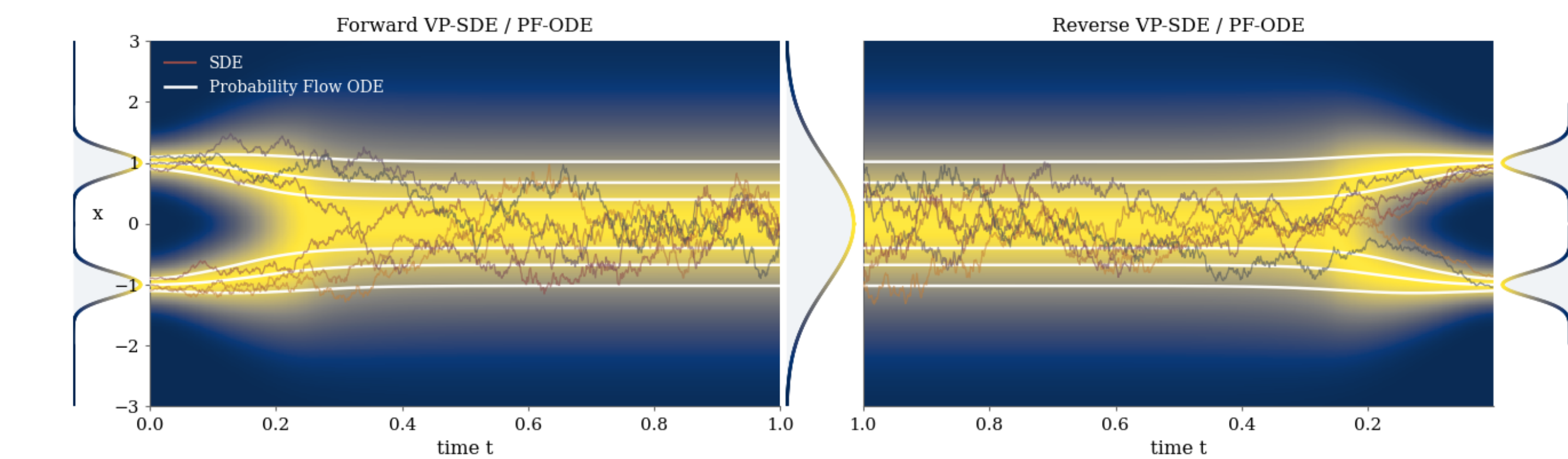
Score-Based Generative Modelling through Stochastic Differential Equations

Georgios Kontos, Zihan Zhu, Elijah Nelson

Motivation

Score-based (SMLD) [3] and **diffusion (DDPM)** [2] models perturb data at **discrete noise levels**, thereby augmenting the training distribution, and learn either the score function or the injected noise. Despite their strong empirical success, these methods require carefully tuned noise schedules. This work [4] generalises them to a **continuous-time framework** by modelling the forward corruption process as a **stochastic differential equation (SDE)** and generation as the corresponding **reverse-time SDE**, with SMLD and DDPM recovered as special cases. The resulting framework enables **high-quality sampling**, **controllable generation**, and **exact likelihood computation** via the **probability flow ODE**.

Score-Based Generative Modelling with SDEs



Forward SDE

$$d\mathbf{x} = \underbrace{f(\mathbf{x}, t)}_{\text{Deterministic drift}} dt + \underbrace{g(t) d\mathbf{w}}_{\text{Stochastic diffusion}}$$

Reverse-Time SDE (Anderson, 1982)

$$d\mathbf{x} = \left[f - g^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt + g d\bar{\mathbf{w}}$$

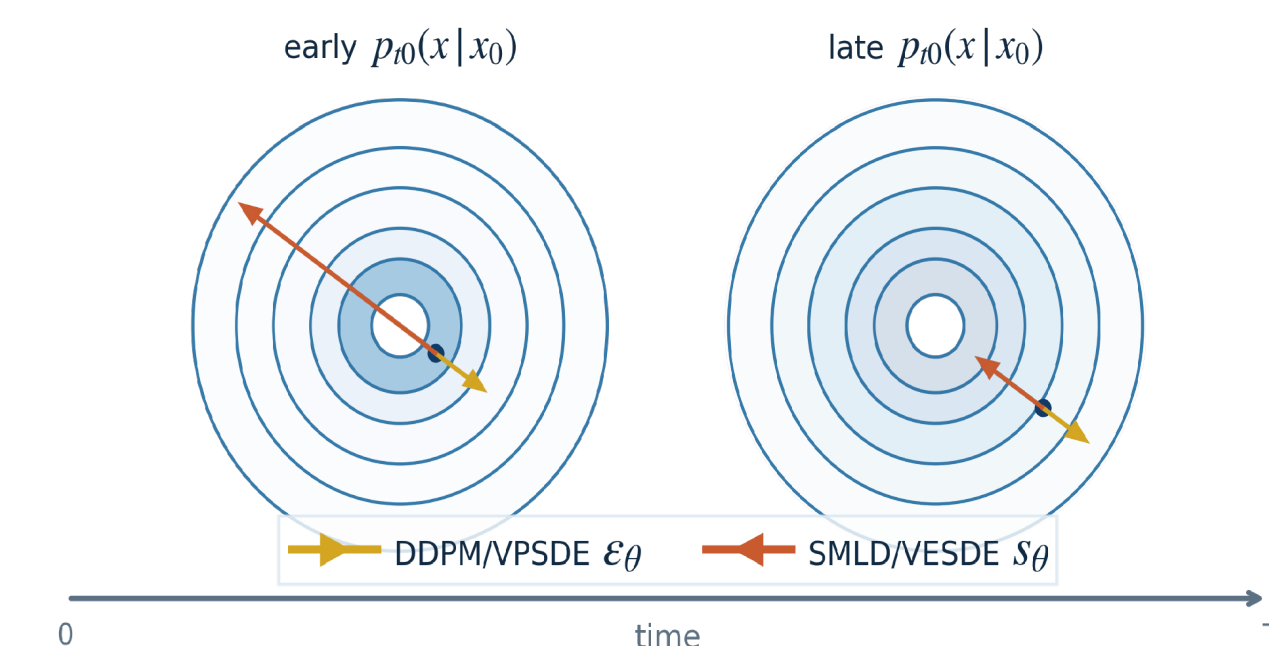
Score function of $p_t(\mathbf{x})$

The SDE is designed to admit: ☒ a tractable prior $p_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for efficient sampling; ☒ a Gaussian transition kernel $p_{0t}(\mathbf{x}(t) | \mathbf{x}(0))$ for closed-form score estimation (affine drift [5]).

Score-Matching Objective & Parameterisations

Train a time-dependent score model $s_{\theta}(\mathbf{x}(t), t) \approx \nabla_{\mathbf{x}} \log p_t$ over $t \sim \mathcal{U}(0, T)$:

$$\begin{aligned} \mathcal{L}(\theta) &= \mathbb{E}_{t, \mathbf{x}(t)} \left[\lambda(t) \left\| s_{\theta} - \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right\|^2 \right] && \text{score matching (intractable)} \\ &\equiv \mathbb{E}_{t, \mathbf{x}(0), \mathbf{x}(t)} \left[\lambda(t) \left\| s_{\theta} - \nabla_{\mathbf{x}} \log p_{0t}(\mathbf{x}(t) | \mathbf{x}(0)) \right\|^2 \right] && \text{denoising score matching (same minimiser)} \\ \nabla_{\mathbf{x}} \log p_{0t} &= -\frac{\mathbf{x}(t) - \mu_t \mathbf{x}(0)}{\sigma_t^2} = -\frac{\boldsymbol{\epsilon}}{\sigma_t} && \mathbf{x}(t) = \mu_t \mathbf{x}(0) + \sigma_t \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \end{aligned}$$



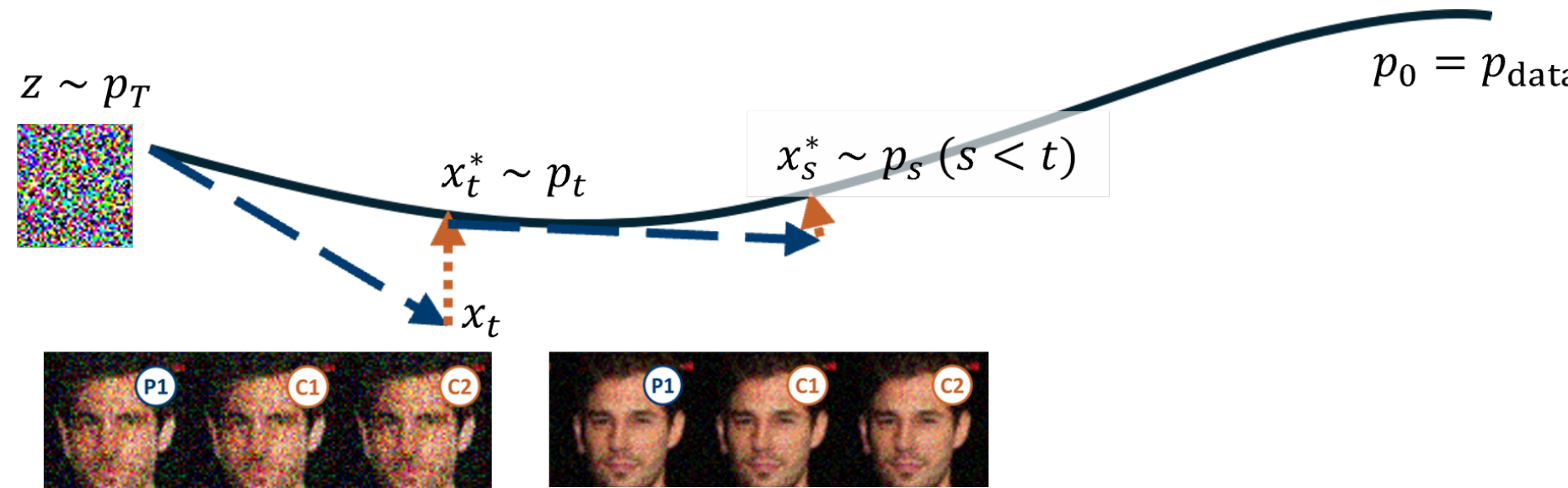
SMLD / VE predict score directly:

$$s_{\theta} \approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x}), \quad \lambda(t) = \sigma^2(t)$$

DDPM / VP predict injected noise:

$$s_{\theta} = -\frac{\boldsymbol{\epsilon}_{\theta}(\mathbf{x}, t)}{\sigma(t)}, \quad \lambda(t) = 1$$

Predictor-Corrector Sampling



Predictor-Corrector (PC) samplers alternate between a **predictor** (numerical SDE solver) and a **corrector** (Langevin MCMC steps to refine the sample towards p_t).

Numerical SDE solvers

- Euler–Maruyama solver (for continuous-time SDEs):

$$\Delta \mathbf{x} \leftarrow \left[f(\mathbf{x}, t) - g^2(t) \underbrace{s_{\theta}(\mathbf{x}, t)}_{\text{score model}} \right] \Delta t + g(t) \mathbf{z}; \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, |\Delta t| \mathbf{I})$$

- Ancestral Sampling: from graphical model $\prod p_{\theta}(\mathbf{x}_{i-1} | \mathbf{x}_i)$, e.g., DDPM discretisation.
- Reverse Diffusion: mirrors the forward SDE's discretisation scheme in reverse time,

$$\mathbf{x}_i = \mathbf{x}_{i+1} - f_{i+1}(\mathbf{x}_{i+1}) + g_{i+1}^2 s_{\theta}(\mathbf{x}_{i+1}, t_{i+1}) + g_{i+1} \mathbf{z}_{i+1}.$$

- Probability Flow: integration of the probability flow ODE backwards in time.

Table 1. FID↓ across predictor types and sampling methods.

	Variance Exploding SDE (SMLD)				Variance Preserving SDE (DDPM)*			
Predictor	P1000	P2000	C2000	PC1000	P1000	P2000	C2000	PC1000
Ancestral sampling	9.879	9.684	—	8.096	5.380	5.379	—	5.282
Reverse diffusion	9.553	9.864	26.905	7.726	5.307	5.308	34.825	5.197
Probability flow	24.883	13.322	—	6.593	8.600	8.597	—	7.076

Sample Quality

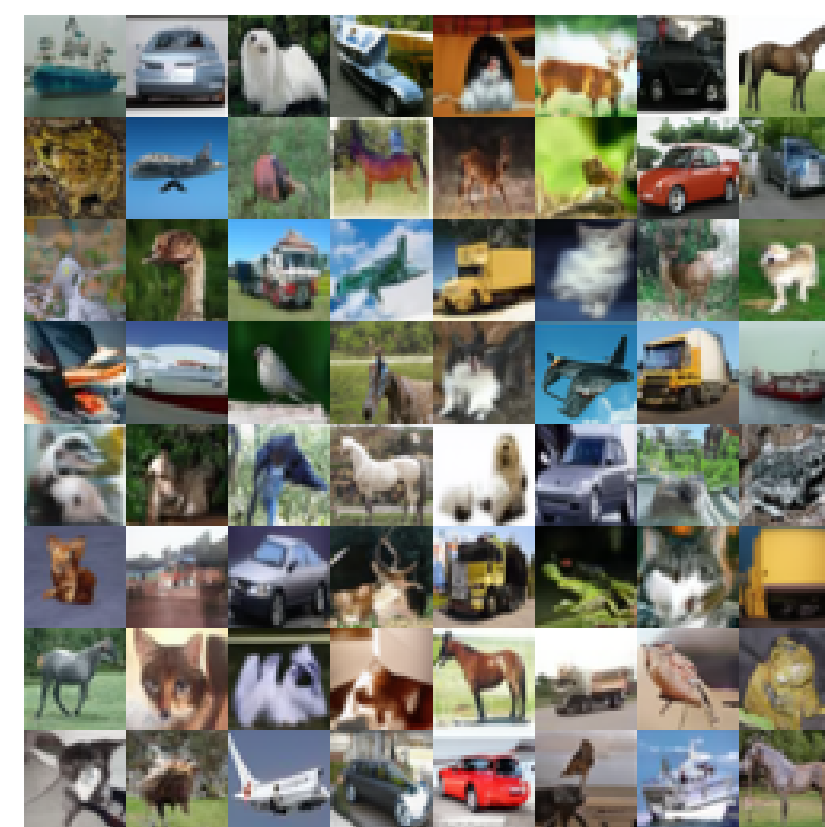


Figure 1. VESDE cont.

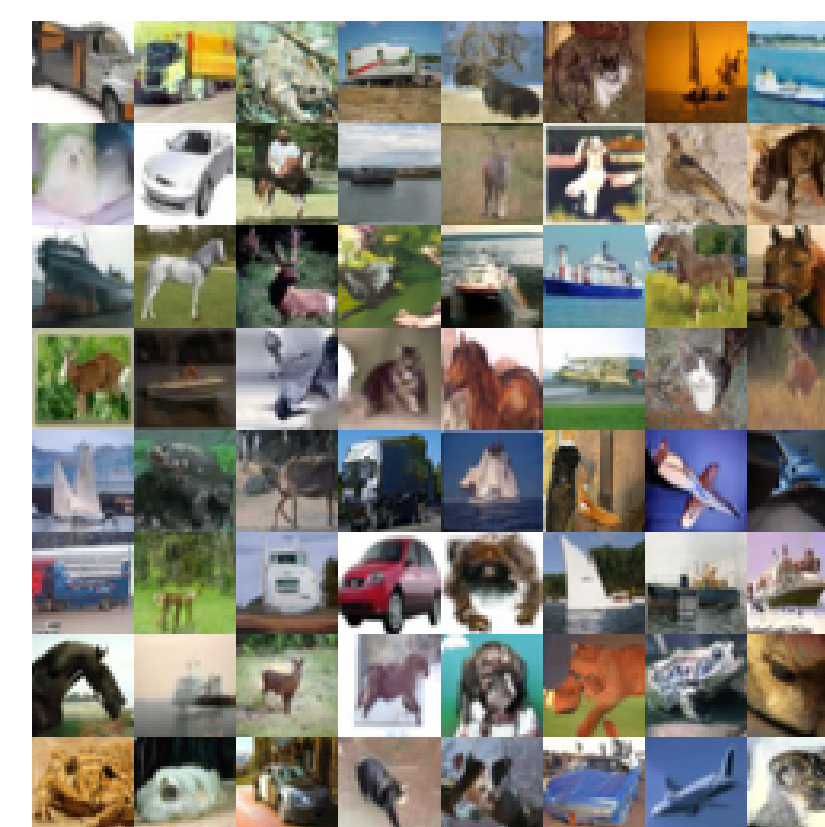


Figure 2. Sub-VP ODE (NCSN++)*.

Table 2. CIFAR-10 sample quality.*

Model	FID↓	IS↑
DDPM++	4.61	9.27
DDPM++ cont. (VP)	5.98	9.51
DDPM++ cont. (sub-VP)	4.77	9.43
DDPM++ cont. (deep, VP)	4.85	9.49
DDPM++ cont. (deep, sub-VP)	4.46	9.25
NCSN++	4.41	9.47
NCSN++ cont. (VE)	4.69	9.71
NCSN++ cont. (deep, VE)	<u>4.45</u>	<u>9.61</u>

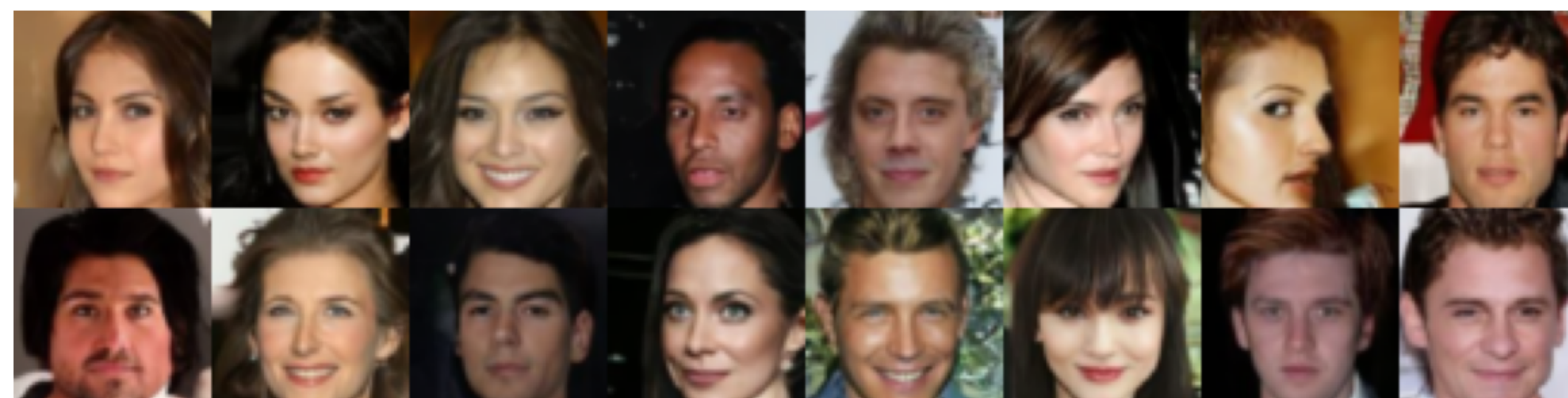


Figure 3. VESDE disc. (CelebA)

Exact Likelihood Computation

The forward SDE admits a deterministic probability flow ODE with identical marginals p_t :

$$d\mathbf{x} = \left[f(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt.$$

Substituting the score model $s_{\theta}(\mathbf{x}, t)$ for $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ yields the neural ODE $\tilde{f}_{\theta}$. The exact log-likelihood then follows from the instantaneous change-of-variables formula [1]:

$$\log p_0(\mathbf{x}) \xleftarrow{\int_0^T \nabla \cdot \tilde{f}_{\theta}(\mathbf{x}(t), t) dt} \log p_T(\mathbf{z})$$

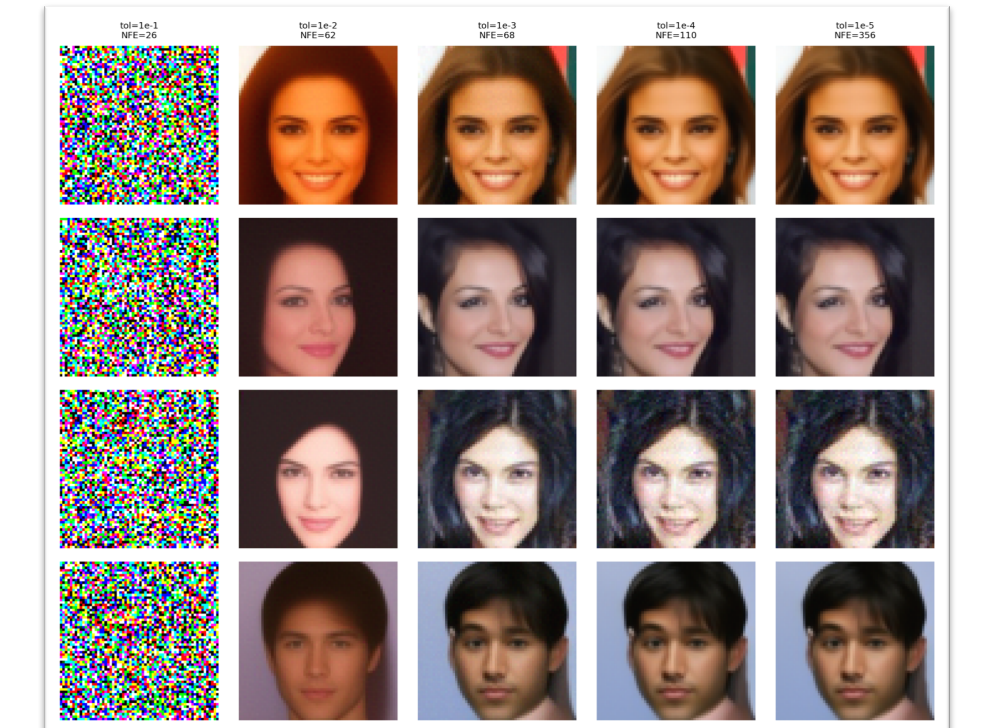
$\nabla \cdot \tilde{f}_{\theta}$ estimated efficiently via the Skilling–Hutchinson trace estimator.

Likelihood Estimation

Table 3. NLLs and FIDs (ODE) on CIFAR-10.*

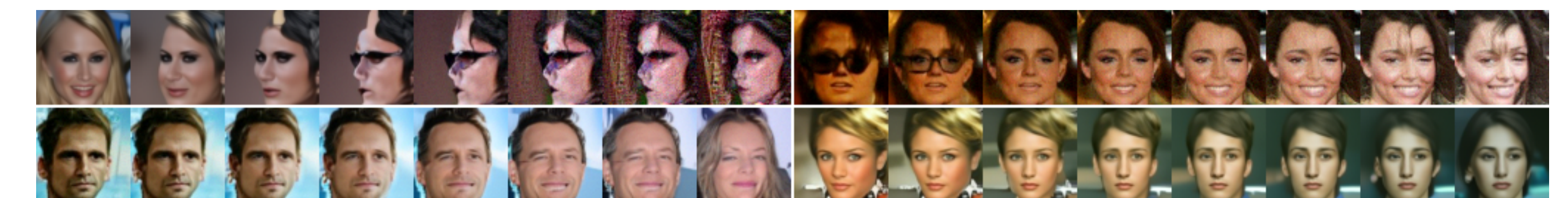
Model	NLL Test↓	FID↓
DDPM	3.26 (3.28)	3.27 (3.37)
DDPM cont. (VP)	3.22 (3.21)	3.59 (3.69)
DDPM cont. (sub-VP)	3.06 (3.05)	3.38 (3.56)
DDPM++ cont. (VP)	3.16 (3.16)	3.70 (3.93)
DDPM++ cont. (sub-VP)	3.02 (3.02)	2.97 (3.16)
DDPM++ cont. (deep, VP)	3.14 (3.13)	2.90 (3.08)
DDPM++ cont. (deep, sub-VP)	2.99 (2.99)	2.69 (2.92)

Figure 4. ODE enables efficient sampling.

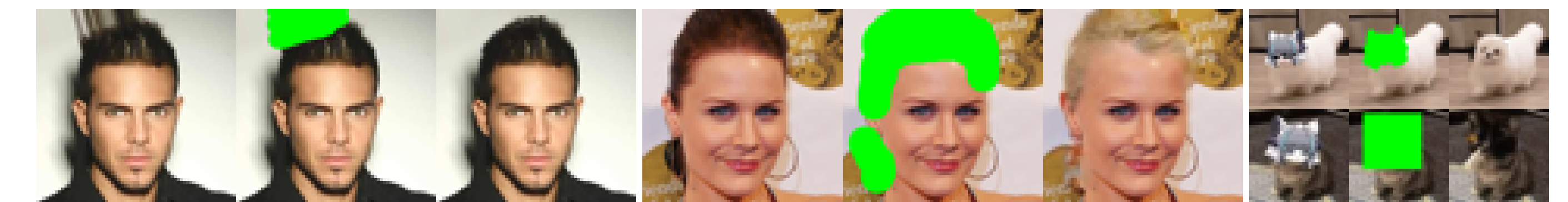


Additional Experiments

Latent Space Interpolation via Probability Flow



Conditional Sampling – Inpainting



References

- [1] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David K Duvenaud. Neural ordinary differential equations. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018.
- [2] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models, 2020.
- [3] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution, 2020.
- [4] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations, 2021.
- [5] Simo Särkkä and Arno Solin. Applied stochastic differential equations. 2019.